

Math Diagnostic
ECON 2010 – Macroeconomics
Spring 2018

Name:

Instructions: *Answer the following math questions to the best of your ability. **If you don't know how to solve a question, that is absolutely fine.** Leave that answer blank. You are **not** being graded on your ability to answer the questions. Instead, this diagnostic is designed to help me, your instructor, understand your problem solving ability with different types of math questions. Knowing your ability will allow me to teach to your level throughout the class.*

Part 1: Computation

1. What is $234 + 169 - 12$?

2. Clara has 8 shelves in her corner store. She has 72 different magazines to display on those 8 shelves. If she wants to display the same number of magazines on each shelf, and display as many magazines as she can, how many can she display on each shelf? Will she have any magazines left over if she displays the same number on each shelf?

3. Chen owns a house valued at \$250,000. He owes property taxes in the amount of 25% to the government on the value of his house. How much in property taxes does Chen owe?

4. Ray ordered a large pizza. After eating, he has three of eight slices remaining. Jesse also ordered a large pizza, but the waiter only cut his pizza into six slices. He ate three of his six slices. If Ray and Jesse combined their leftover slices, what fraction of a total large pizza will they have?

5. What is 3^3 ?

Part 2: Geometry

6. What is the slope of the line that passes through the points (3,5) and (6,10) on the cartesian plane (i.e. the x,y plane)?

Part 3: Algebra

7. Consider the following equation: $6 = \frac{2}{3}x - 2$. Solve for x .

8. Juan has opened a savings account and deposited \$100.00 (the principal). The account earns 10% interest compounded annually (i.e. once per year). How much money will Juan have in 3 years? Use the following formula for compound interest to calculate the amount:

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

where A represents the final amount, P represents the principal, r represents the rate of interest (expressed as a decimal), n is the number of times per year that interest is compounded, and t is the number of years.

Part 4: Probability

9. A card is drawn randomly from a standard deck of 52 playing cards (no jokers). What is the probability (expressed as a fraction) that you will draw a queen? If you replace that queen in the deck and draw a second card, what is the probability that you will draw two queens in a row?

Part 5: Calculus and Higher Level Math

10. Take the derivative of $y = (3x + 1)^2$ with respect to x .

11. Prove that $\sqrt{2}$ is irrational.